



General purpose PLL

Team X

Yutaka KOTANI

Jay

Teddy

Butter cutter

Agenda

1 Project overview

2 Block diagram

3 Specification

4 Circuit design

5 Evaluation

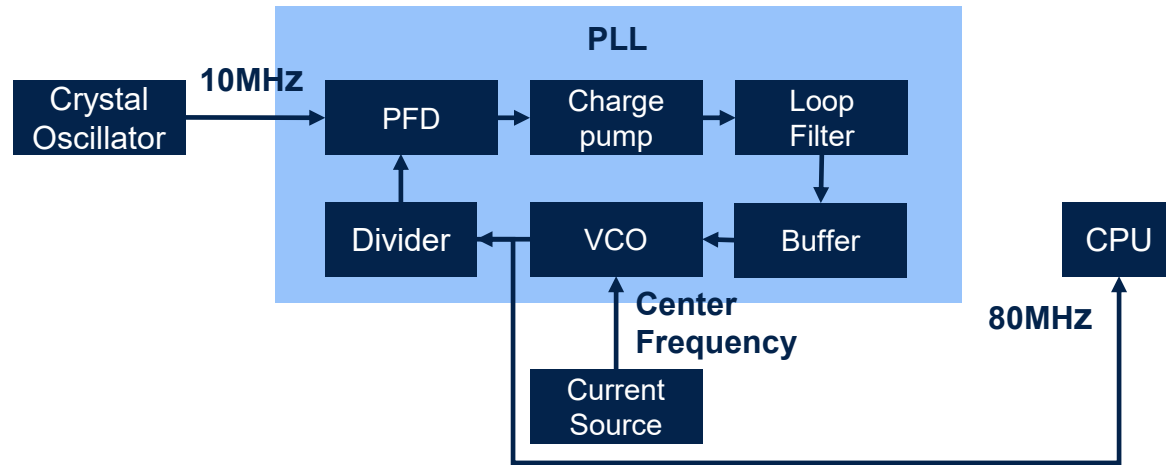
6 Layout

7 PEX & Validation

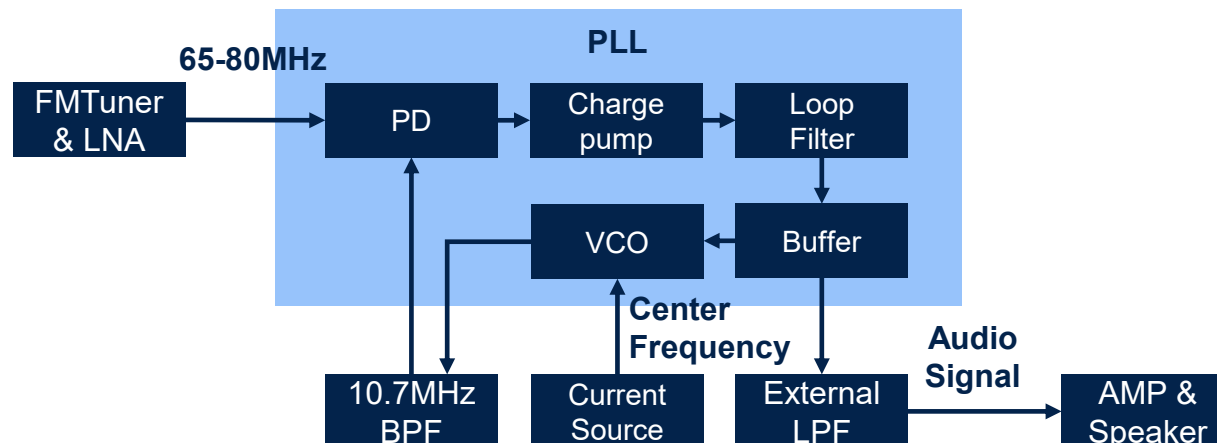
8 Schedule

Project overview

Clock multiplier mode

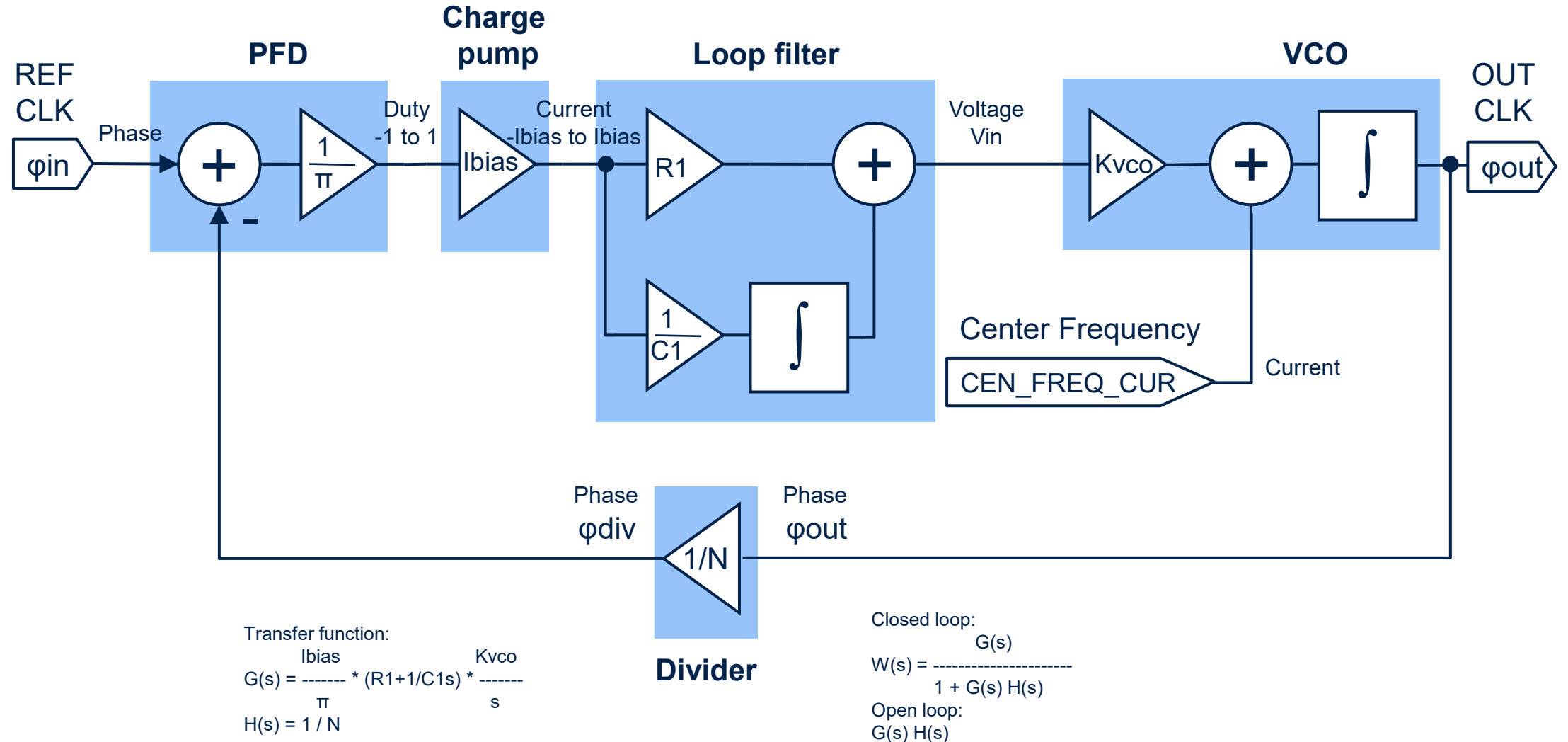


FM demodulator mode

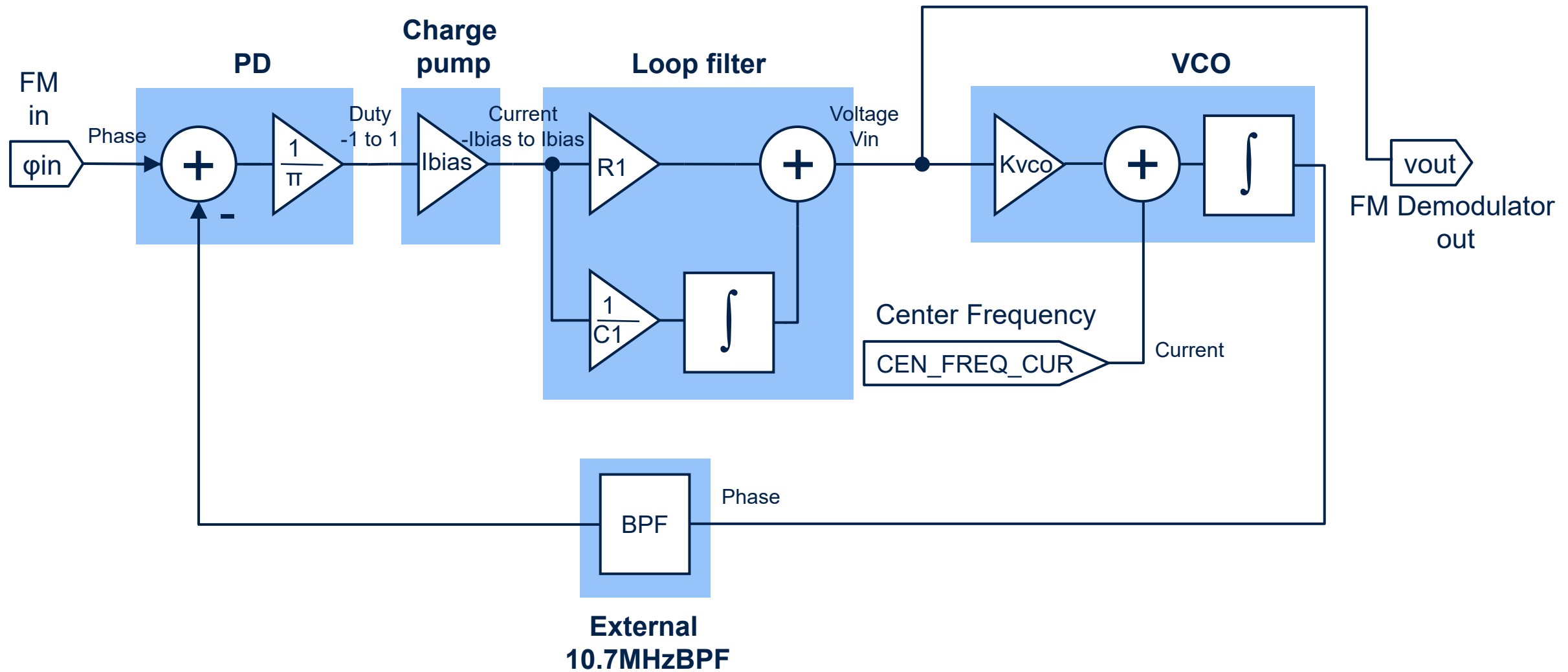


- Team member and role
 - Yutaka KOTANI – Team leader, VCO
 - Jay - PFD
 - Teddy - Divider
 - Butter cutter - PFD
- PLL for general purpose
 - High speed CPU clock generation
 - FM radio demodulator
- PLL has two mode
 - Clock multiplier mode
 - FM demodulator mode

Block diagram - Clock multiplier mode



Block diagram – FM demodulator mode

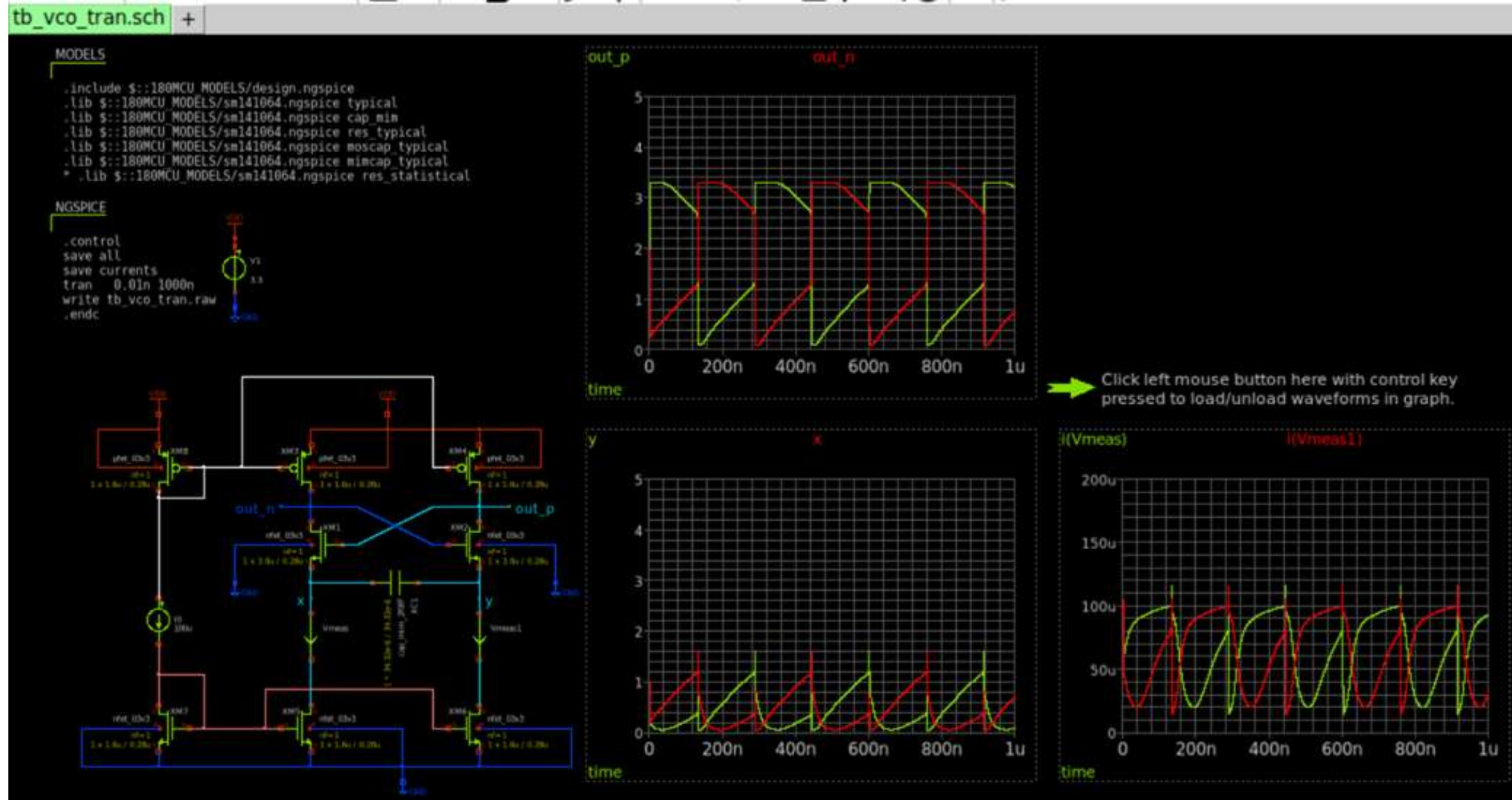


Specification (T.B.D)

Target Spec	Min	Typ	Max
Pin count	9		
Estimated Area	999um2		
Vdd voltage [V]	3	3.3	3.6
pll_in voltage [V]	0.8	-	2.5
pll_out voltage [V]	0		vdd
Clock multiplier mode	Min	Typ	Max
pll_in frequency [MHz]	9.9	10	10.1
pll_out frequency [MHz]	79.9	80	80.1
Divider ratio		1/8	
Cen_freq_cur [uA]	90	100	110
FM demodulator mode	Min	Typ	Max
pll_in frequency [MHz]	76	-	90
dmod_out voltage [V]	0	-	vdd
Cen_freq_cur [uA]	10	-	110

Pin	Pin name	Notes
1	VDD	3.3V
2	GND	
3	PLL_IN	Reference clk input
4	PLL_OUT	Clock multiplier output
5	CEN_FREQ_CUR	Center frequency bias current
6	MODE_SEL	Hi: Use FM demodulator Low: Use Clock multiplier
7	EXT_FIL_OUT	To Band pass filter 10.7MHz
8	EXT_FIL_IN	From Band pass filter 10.7MHz
9	DMOD_OUT	FM demodulator output

Circuit design - VCO



Circuit design – module name

Evaluation (T.B.D)

- Put evaluation result here
- Insert graph to left

Layout (T.B.D)

PEX & Validation (T.B.D)

Schedule

Week	Date	Event	Track Details
Week 23	June 5	Team Formation Deadline	
Week 24	June 12	Project Proposal Review	
Week 27	July 3	Schematic Review	
Week 28	July 10	Simulation Review (blocks)	
Week 29	July 17	Simulation Review (top level)	
		READiness Check & Go/No-go Decision	
Week 33	Aug 14	Layout Review (blocks)	
Week 34	Aug 21	Layout Review (top level)	
Week 35	Aug. 28	Verification	
		Final Chip Review	
		Final Submission	
		Upload of Chip Test Results	